

Bayesian Regression

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```
set.seed(42)
library(brms)
```

```
## Loading required package: Rcpp
```

```
## Loading 'brms' package (version 2.23.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').
```

```
##
## Attaching package: 'brms'
```

```
## The following object is masked from 'package:stats':
##
##      ar
```

```
library(ggplot2)
```

```
avo <- read.csv("spoilage_avo.csv")
head(avo)
```

##	X	year	Locations.State.Code	Location.State.Abbreviation	Location.County.Code
## 1	0	2002	6	CA	73
## 2	1	2002	6	CA	73
## 3	2	2002	6	CA	73
## 4	3	2002	6	CA	73
## 5	4	2002	6	CA	73
## 6	5	2002	6	CA	79
##	Location.County.Name	Commodity.Code	Commodity.Name	Insurance.Plan.Code	
## 1	San Diego	19	AVOCADOS	46	
## 2	San Diego	19	AVOCADOS	46	
## 3	San Diego	19	AVOCADOS	46	
## 4	San Diego	19	AVOCADOS	46	
## 5	San Diego	19	AVOCADOS	46	
## 6	San Luis Obispo	19	AVOCADOS	46	
##	Insurance.Plan.Abbreviation	Stage.Code	Damage.Cause.Code		
## 1	ARC	H	12		
## 2	ARC	H	43		
## 3	ARC	H	61		
## 4	ARC	H	91		
## 5	ARC	UH	91		
## 6	ARC	H	44		
##	Damage.Cause.Description	Month.of.Loss	Month.of.Loss.Abbreviation		
## 1	Heat	12	DEC		
## 2	Cold Winter	12	DEC		
## 3	Wind/Excess Wind	7	JUL		
## 4	Fire	2	FEB		
## 5	Fire	2	FEB		
## 6	Cold Wet Weather	3	MAR		
##	Determined.Acres	Indemnity.Amount	month_year	Average.Temperature	Total.CDD
## 1	5.000	3578.0	2002-12-01	44.0	0
## 2	3.200	1322.0	2002-12-01	44.0	0
## 3	2.400	1185.2	2002-07-01	77.4	261
## 4	0.225	85.5	2002-02-01	48.2	0
## 5	0.585	403.2	2002-02-01	48.2	0
## 6	41.080	50117.8	2002-03-01	48.5	0
##	Total.HDD	Total.Precipitation	Avg.PDSI		
## 1	536	8.14	1.60		
## 2	536	8.14	1.60		
## 3	0	0.06	-3.07		
## 4	388	1.45	-2.24		
## 5	388	1.45	-2.24		
## 6	452	1.91	-2.48		

```
# drop state and commodity columns (already filtered for CA and avocados)
avo$Locations.State.Code <- NULL
avo$Locations.State.Abbreviation <- NULL
avo$Commodity.Code <- NULL
avo$Commodity.Name <- NULL

# take log transform of indemnity amt b/c values represent money amount so they're positive
avo$Indemnity.Amount <- as.numeric(gsub(",", "", avo$Indemnity.Amount))
avo$log_indemnity <- ifelse(avo$Indemnity.Amount > 0, log(avo$Indemnity.Amount), NA)
```

```
## Warning in log(avo$Indemnity.Amount): NaNs produced
```

```
# drop rows where vals are NA
keep_cols <- c("log_indemnity", "Determined.Acres", "Average.Temperature",
              "Total.CDD", "Total.HDD", "Total.Precipitation", "Avg.PDSI",
              "Damage.Cause.Description", "Stage.Code", "Month.of.Loss",
              "Location.County.Name")
avo_clean <- avo[complete.cases(avo[, keep_cols]), keep_cols]

# encode categorical variables as factors for random intercepts in brms
avo_clean$Damage.Cause.Description <- as.factor(avo_clean$Damage.Cause.Description)
avo_clean$Stage.Code <- as.factor(avo_clean$Stage.Code)
avo_clean$Month.of.Loss <- as.factor(avo_clean$Month.of.Loss)
avo_clean$Location.County.Name <- as.factor(avo_clean$Location.County.Name)

# scale numeric vals
numeric_covs <- c("Determined.Acres", "Average.Temperature", "Total.CDD", "Total.HDD",
                  "Total.Precipitation", "Avg.PDSI")
avo_scaled <- avo_clean
avo_scaled[, numeric_covs] <- scale(avo_clean[, numeric_covs])
head(avo_scaled)
```

##	log_indemnity	Determined.Acres	Average.Temperature	Total.CDD	Total.HDD
## 1	8.182559	-0.2910691	-1.0300872	-0.6086695	1.2162417
## 2	7.186901	-0.2996502	-1.0300872	-0.6086695	1.2162417
## 3	7.077667	-0.3034640	1.9279345	2.2424418	-1.4629328
## 4	4.448516	-0.3138328	-0.6581204	-0.6086695	0.4764696
## 5	5.999433	-0.3121166	-0.6581204	-0.6086695	0.4764696
## 6	10.822132	-0.1190663	-0.6315513	-0.6086695	0.7963711

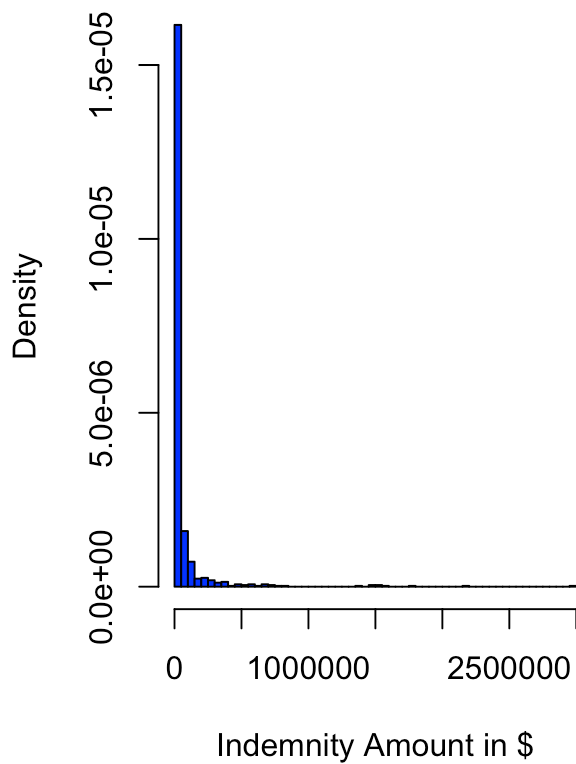
##	Total.Precipitation	Avg.PDSI	Damage.Cause.Description	Stage.Code
## 1	2.6276587	1.1559219	Heat	H
## 2	2.6276587	1.1559219	Cold Winter	H
## 3	-0.9646256	-0.4521915	Wind/Excess Wind	H
## 4	-0.3466460	-0.1663812	Fire	H
## 5	-0.3466460	-0.1663812	Fire	UH
## 6	-0.1421347	-0.2490251	Cold Wet Weather	H

##	Month.of.Loss	Location.County.Name
## 1	12	San Diego
## 2	12	San Diego
## 3	7	San Diego
## 4	2	San Diego
## 5	2	San Diego
## 6	3	San Luis Obispo

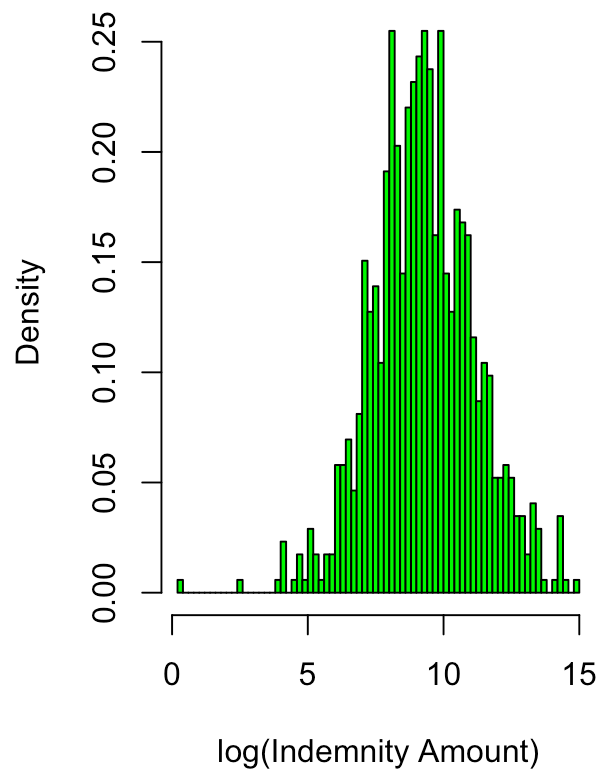
```
# indemnity amount before/after log (dist becomes normal)
par(mfrow = c(1, 2))
hist(exp(avo_clean$log_indemnity), breaks = 60, probability = TRUE,
     col = "blue",
     main = "Indemnity Amount", xlab = "Indemnity Amount in $")

hist(avo_clean$log_indemnity, breaks = 60, probability = TRUE,
     col = "green",
     main = "log(Indemnity Amount)", xlab = "log(Indemnity Amount)")
```

Indemnity Amount



log(Indemnity Amount)



```
par(mfrow = c(1, 1))
```

```
# brms for bayesian reg
```

```
default_prior(log_indemnity ~ Determined.Acres + Average.Temperature +  
              Total.CDD + Total.HDD + Total.Precipitation + Avg.PDSI +  
              (1 | Damage.Cause.Description) + (1 | Stage.Code) +  
              (1 | Month.of.Loss) + (1 | Location.County.Name), data = avo_scaled,  
              family = gaussian())
```

[illegible]

```

manual_prior_brms <- c(
  prior(student_t(3, 0, 10), class = "Intercept"),
  prior(normal(0, 10), class = "b", coef = "Determined.Acres"),
  prior(normal(0, 10), class = "b", coef = "Average.Temperature"),
  prior(normal(0, 10), class = "b", coef = "Total.CDD"),
  prior(normal(0, 10), class = "b", coef = "Total.HDD"),
  prior(normal(0, 10), class = "b", coef = "Total.Precipitation"),
  prior(normal(0, 10), class = "b", coef = "Avg.PDSI"))

avo_brm <- brm(
  log_indemnity ~ Determined.Acres + Average.Temperature +
    Total.CDD + Total.HDD + Total.Precipitation + Avg.PDSI +
    (1 | Damage.Cause.Description) + (1 | Stage.Code) +
    (1 | Month.of.Loss) + (1 | Location.County.Name), family = gaussian(),
  data = avo_scaled,
  chains = 4,
  cores = 2,
  iter = 2000,
  warmup = 400,
  thin = 2,
  prior = manual_prior_brms,
  control = list(adapt_delta = 0.95))

```

```
## Compiling Stan program...
```

```
## Trying to compile a simple C file
```

```
## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 15.0.0 (clang-1500.1.0.2.5)'
## using SDK: 'MacOSX14.2.sdk'
## clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -D
NDEBUG -I"/Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/Rcpp/in
clude/" -I"/Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEig
en/include/" -I"/Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/Rc
ppEigen/include/unsupported" -I"/Library/Frameworks/R.framework/Versions/4.5-arm64/Reso
urces/library/BH/include" -I"/Library/Frameworks/R.framework/Versions/4.5-arm64/Resource
s/library/StanHeaders/include/src/" -I"/Library/Frameworks/R.framework/Versions/4.5-arm
64/Resources/library/StanHeaders/include/" -I"/Library/Frameworks/R.framework/Versions/
4.5-arm64/Resources/library/RcppParallel/include/" -I"/Library/Frameworks/R.framework/V
ersions/4.5-arm64/Resources/library/rstan/include" -DEIGEN_NO_DEBUG -DBOOST_DISABLE_ASS
ERTS -DBOOST_PENDING_INTEGER_LOG2_HPP -DSTAN_THREADS -DUSE_STANC3 -DSTRICT_R_HEADERS
-DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION -D_HAS_AUTO_PTR_ETC=0 -include '/Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp' -D_REENTRANT -DRCPP_PARALLEL_USE_TBB=1 -I/opt/R/arm64/include -fPIC -falign-functions=64 -Wall -g -O2 -c foo.c -o foo.o
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
## /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: 'cmath' file not found
## #include <cmath>
##          ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
```

```
## Start sampling
```

```
##           8.06 seconds (Sampling)
## Chain 1:           11.631 seconds (Total)
## Chain 1:
## Chain 2: Iteration: 1200 / 2000 [ 60%] (Sampling)
```

```
## Warning: There were 39 divergent transitions after warmup. See
## https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
## to find out why this is a problem and how to eliminate them.
```

```
## Warning: Examine the pairs() plot to diagnose sampling problems
```

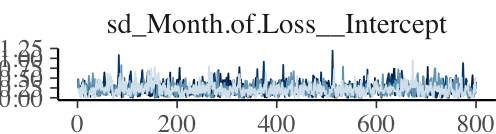
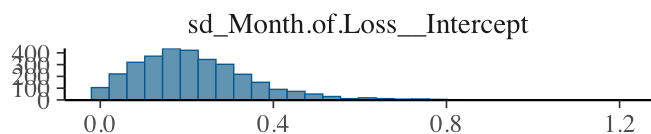
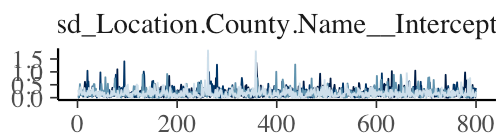
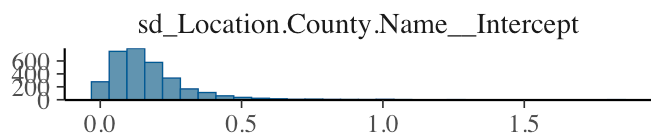
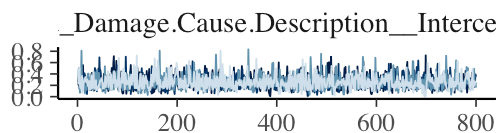
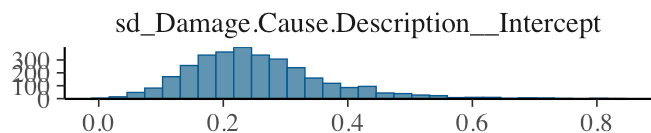
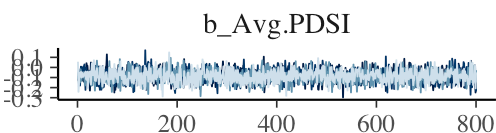
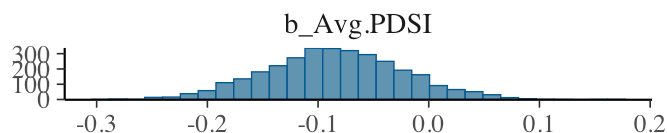
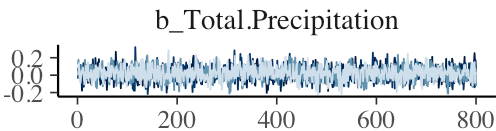
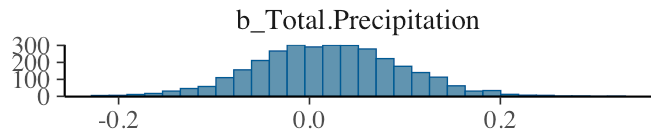
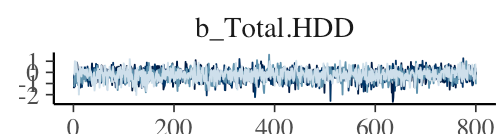
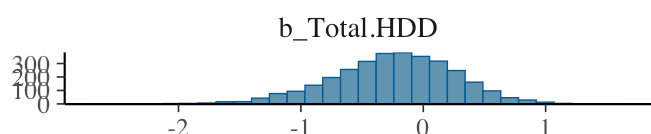
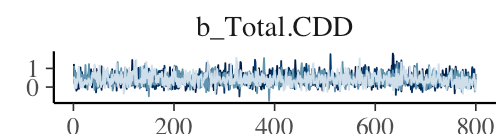
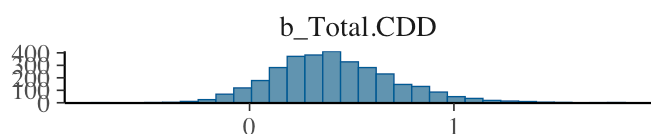
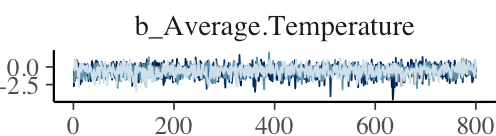
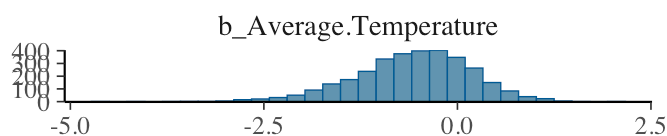
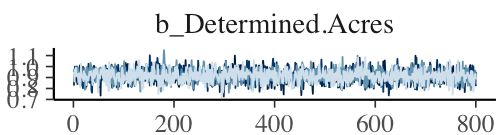
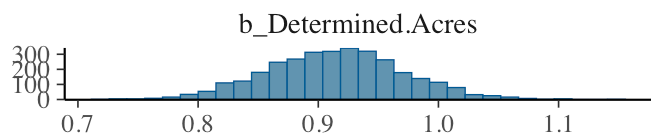
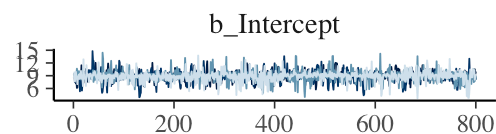
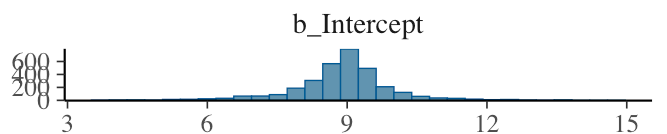
```
summary(avo_brm)
```



```
## Warning: There were 39 divergent transitions after warmup. Increasing
## adapt_delta above 0.95 may help. See
## http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
```

```
## pitation + Avg.PDSI + (1 | Damage.Cause.Description) + (1 | Stage.Code) + (1 | Month.
of.Loss) + (1 | Location.County.Name)
## Data: avo_scaled (Number of observations: 863)
## Draws: 4 chains, each with iter = 2000; warmup = 400; thin = 2;
## total post-warmup draws = 3200
##
## Multilevel Hyperparameters:
## ~Damage.Cause.Description (Number of levels: 23)
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept) 0.26 0.11 0.08 0.52 1.00 1898 2213
##
## ~Location.County.Name (Number of levels: 5)
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept) 0.17 0.15 0.01 0.55 1.00 1617 1642
##
## ~Month.of.Loss (Number of levels: 12)
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept) 0.22 0.14 0.02 0.54 1.00 987 1634
##
## ~Stage.Code (Number of levels: 2)
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept) 1.32 1.12 0.20 4.37 1.00 2190 2442
##
## Regression Coefficients:
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept 8.89 1.13 6.24 11.36 1.00 2141 1777
## Determined.Acres 0.91 0.06 0.80 1.02 1.00 2654 2733
## Average.Temperature -0.58 0.75 -2.20 0.74 1.00 2202 2640
## Total.CDD 0.42 0.30 -0.11 1.07 1.00 1912 2793
## Total.HDD -0.25 0.51 -1.30 0.66 1.00 2397 2711
## Total.Precipitation 0.02 0.08 -0.13 0.17 1.00 2340 2592
## Avg.PDSI -0.09 0.06 -0.21 0.04 1.00 2084 2737
##
## Further Distributional Parameters:
## Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma 1.57 0.04 1.50 1.65 1.00 2578 2266
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(avo_brm)
```

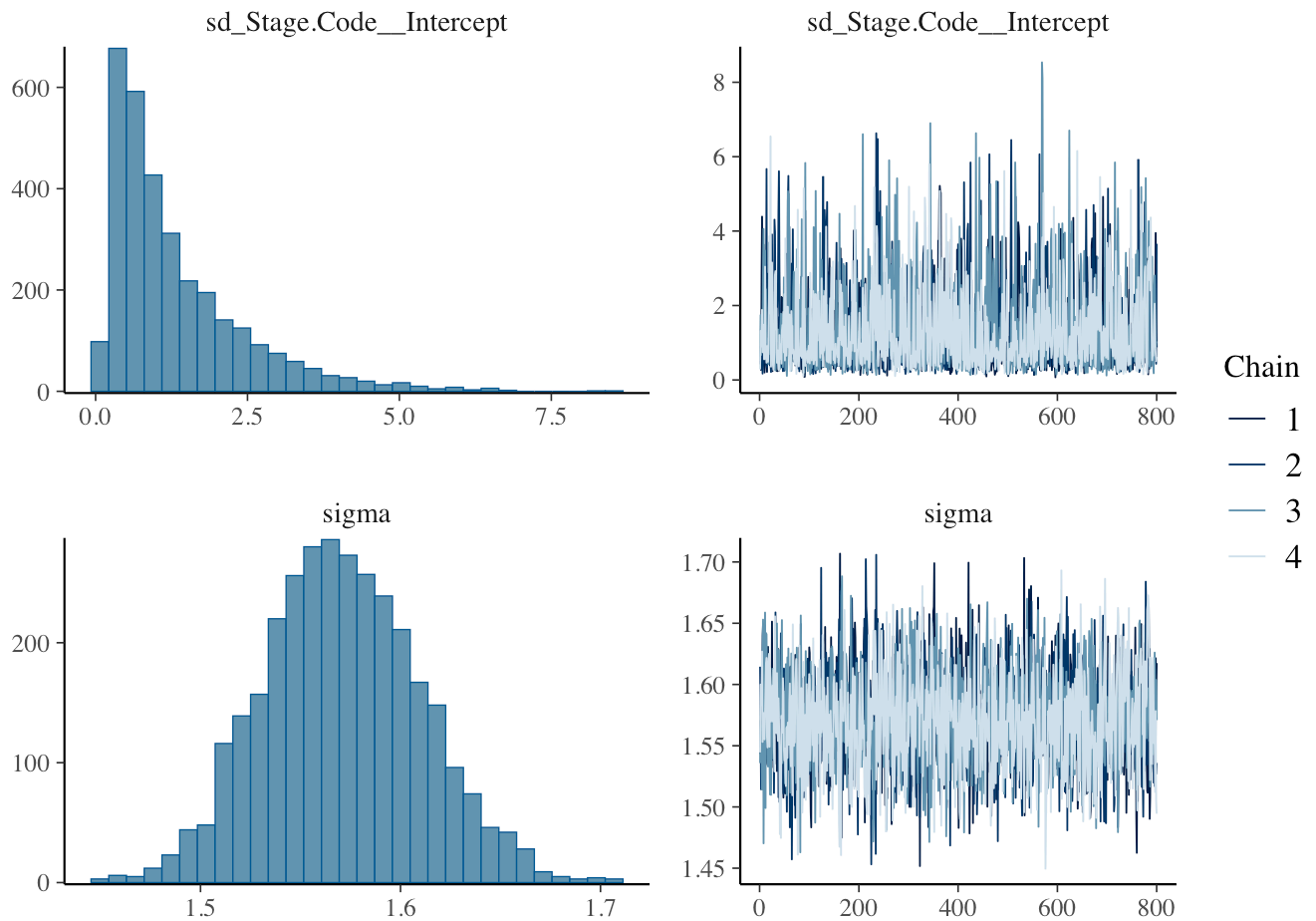


Chain

— 1
— 2
— 3
— 4

Chain

— 1
— 2
— 3
— 4

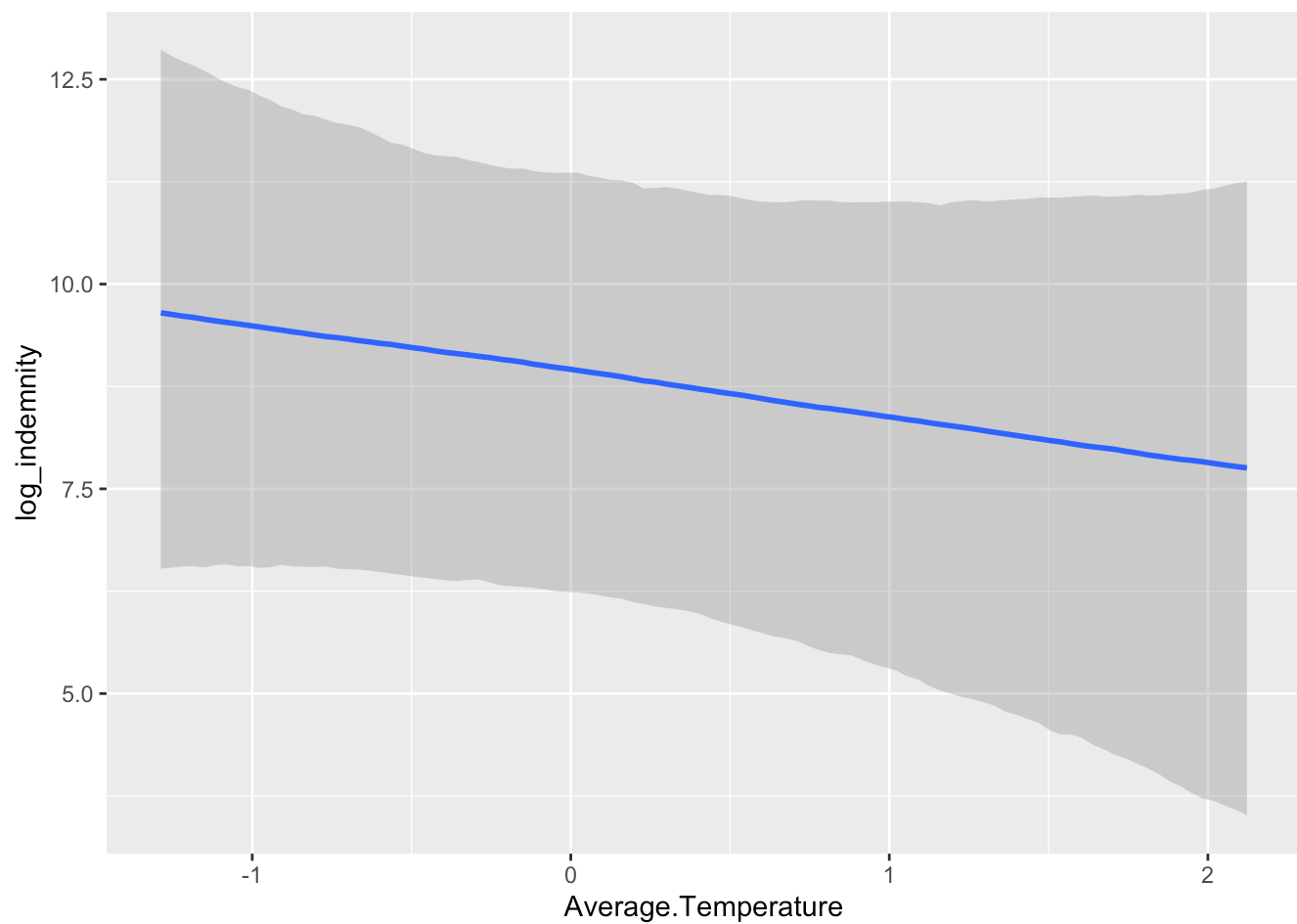


```
bayes_R2(avo_brm)
```

```
##      Estimate Est.Error    Q2.5    Q97.5
## R2  0.3123761 0.02217592 0.266814 0.3550692
```

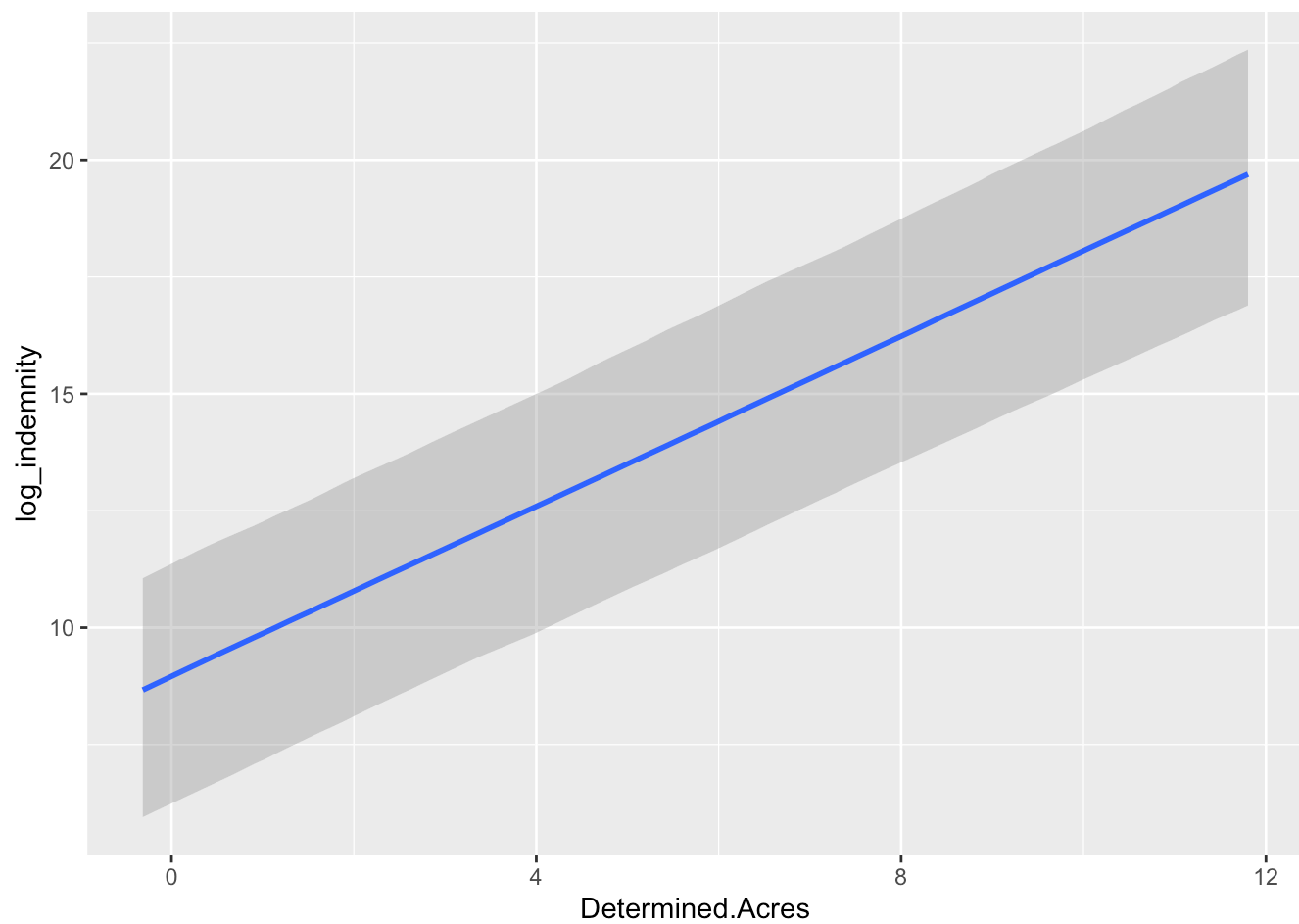
```
# plot the way that log indemnity changes w/ most significant features
plot(conditional_effects(avo_brm, "Average.Temperature"))
```

```
## Ignoring unknown labels:
## • fill : "NA"
## • colour : "NA"
## Ignoring unknown labels:
## • fill : "NA"
## • colour : "NA"
```



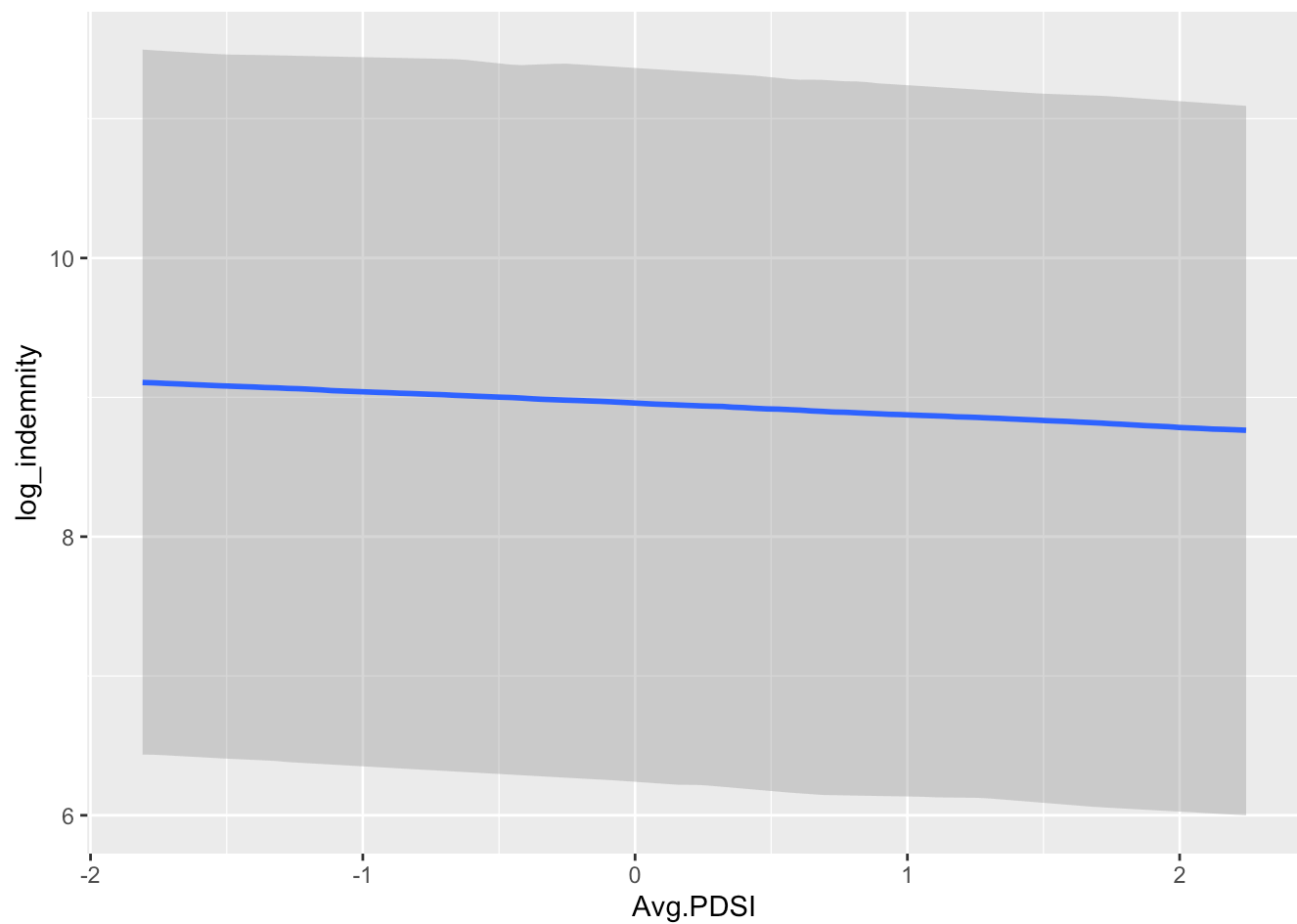
```
plot(conditional_effects(avo_brm, "Determined.Acres"))
```

```
## Ignoring unknown labels:  
## • fill : "NA"  
## • colour : "NA"  
## Ignoring unknown labels:  
## • fill : "NA"  
## • colour : "NA"
```



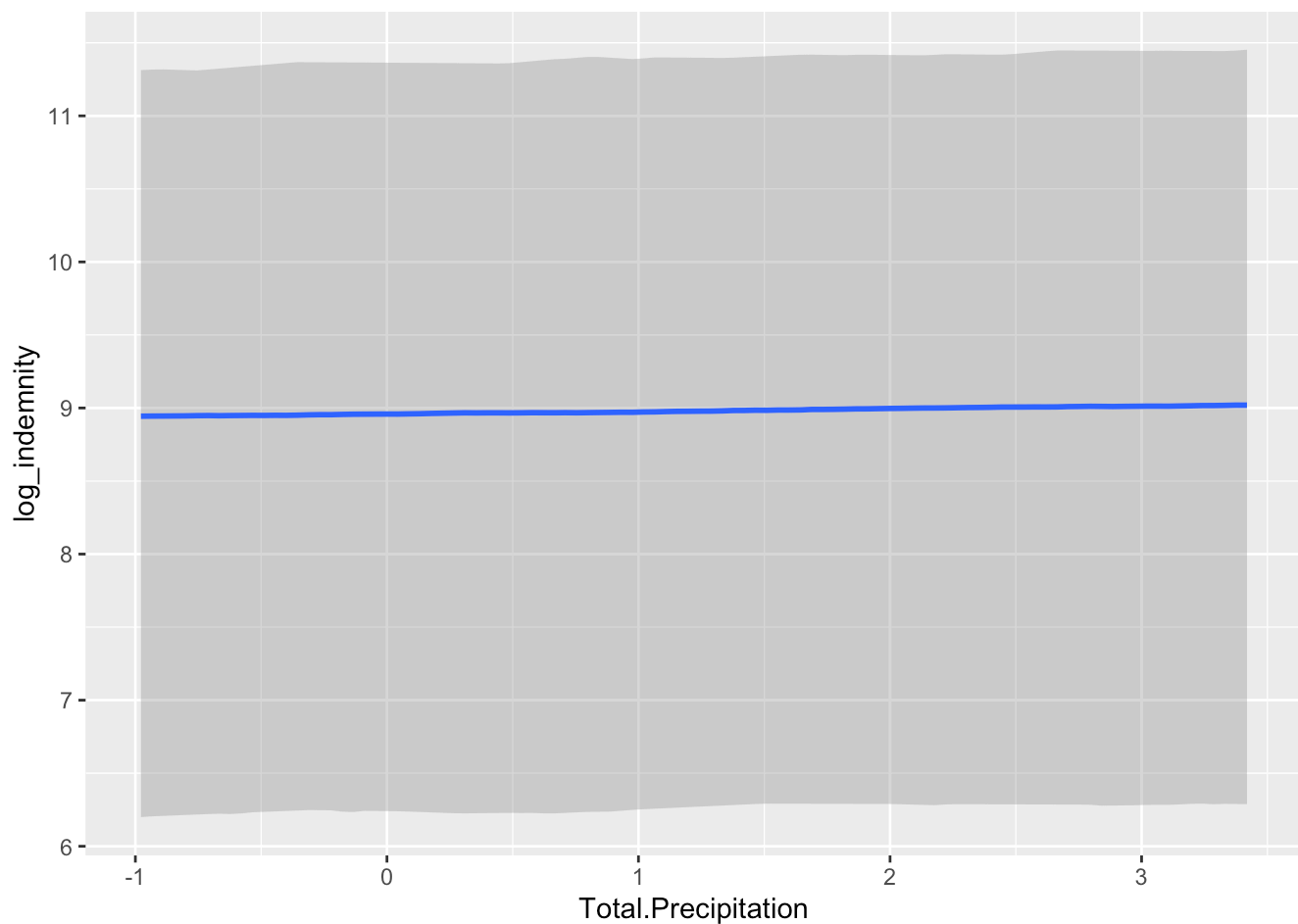
```
plot(conditional_effects(avo_brm, "Avg.PDSI"))
```

```
## Ignoring unknown labels:  
## • fill : "NA"  
## • colour : "NA"  
## Ignoring unknown labels:  
## • fill : "NA"  
## • colour : "NA"
```



```
plot(conditional_effects(avo_brm, "Total.Precipitation"))
```

```
## Ignoring unknown labels:  
## • fill : "NA"  
## • colour : "NA"  
## Ignoring unknown labels:  
## • fill : "NA"  
## • colour : "NA"
```



```
# posterior pred mean & RMSE
post_preds_brm <- posterior_predict(avo_brm)
brm_pred_means <- colMeans(post_preds_brm)

y <- avo_scaled$log_indemnity
brm_rmse <- sqrt(mean((y - brm_pred_means)^2))
cat("BRMS RMSE:", round(brm_rmse, 4))
```

```
## BRMS RMSE: 1.5475
```

```
plot(brm_pred_means, y, xlab = "BRMS Posterior Predictive Mean log(Indemnity Amount)",
     ylab = "Observed log(Indemnity Amount)",
     main = "BRMS: Posterior Predictive vs. Observed",
     pch = 16, col = "red")
abline(0, 1, col = "red", lwd = 2)
```

BRMS: Posterior Predictive vs. Observed

